

1b

no. of banks = 26

stem	Leaves
1	5 5 8 8
2	0 0 0 0 1 2 2 5 5 5 5 5 8 8 9
3	4 5 6 7 7
4	1 2

- i) Two information obtained from stem and leaf display are:
- how data are distributed
 - central tendency

ii) no. of items before 30 = 19
percentage " " = $\frac{19}{26} \times 100\%$

= 73.08%

k:

$1 + 3.222$
 $\log_{10}(N)$

iii) $K = 1 - 3.222 \log_{10}(26) = -14.61$

$w = \frac{42-15}{14.61} = 1.85$

$k = \frac{42-15}{14}$
 1.93

Choose class interval difference of 5.

x	f
10-15	2
15-20	2
20-25	7
25-30	8
30-35	1
35-40	4
40-45	2

1 a

Marks = x

no. of students = f

①

Marks less than	f	$\text{cum } f$
20	4	4
30	6	10
40	10	20
50	20	40
60	18	58
70	2	60

Marks more than	f
10	60
20	56
30	50
40	40
50	20
60	2
70	

from graph, median = 45

2a

income (000) = x
no. of persons = f

x	f	cf
0-20	6	6
20-40	25	31
40-60	35	66
60-80	25	91
80-100	9	100
	$N=100$	

① middle 30% (P_{35} , P_{65})

position of P_{35}

$$= \left(35 \times \frac{N}{100} \right)^{\text{th term}}$$

$$= \left(\frac{35 \times 100}{100} \right)^{\text{th}}$$

$$= 35^{\text{th}}$$

$$= (40-60)$$

$$\therefore P_{35} = 40 + \left(\frac{(35 - 31)}{35} \times 20 \right)$$

$$= 42.29$$

$$\begin{aligned}\text{position of } P_{65} &= \left(65 \times \frac{100}{100}\right)^{\text{th}} \text{ term} \\ &= 65^{\text{th}} \\ &= (90-60)\end{aligned}$$

$$\begin{aligned}P_{65} &= 40 + \left(\frac{65-31}{35} \times 200\right) \\ &= 59.43\end{aligned}$$

$\therefore$ income limits of middle 30% workers is.

$$\begin{aligned}&(\underline{42.29 - 59.43}) \\ 42.29 \times 100 &= 42290 \text{ Rs.}, 59430 \text{ Rs.}\end{aligned}$$

ii) highest income of poorest 25% = P_{25}

$$\begin{aligned}P_{25} &= 25 \times \frac{100}{100} = 25^{\text{th}} \text{ term} \\ &= (20-90)\end{aligned}$$

$$\begin{aligned}P_{25} &= 20 + \left(\frac{25-6}{25} \times 20\right) \\ &= 35.20\end{aligned}$$

$\therefore$ highest income of poorest = 35200 Rs.

2b
variable: x
frequency: f

let, missing frequencies
are x_1 and y .

x	f	cf
0-10	5	5
10-20	7	12
20-30	x_1	$12+x_1$
30-40	15	$27+x_1$
40-50	25	$52+x_1$
50-60	y	$52+x_1+y$
60-70	10	$62+x_1+y$
	$N=100$	

$$Md = 36 = (30-40)$$

$$N = 62+x_1+y$$

$$x_1+y = 38 \quad \text{--- (1)}$$

$$Md = 30 + \frac{50 - (27+x_1)}{15} \times 10$$

$$36 = 30 + \frac{38-x_1}{15} \times 10$$

$$x_1 = 29$$

$$y = 38 - 29 = 9$$

2b

∴ missing frequencies are 29 and 9
of (20-30) and (50-60) respectively.

3a

for model A

Life (No. of years)	x	f	fx	fx^2
0-2	1	5	5	5
2-4	3	16	48	144
4-6	5	13	65	325
6-8	7	7	49	343
8-10	9	5	45	405
10-12	11	4	44	484
		$N = 50$	$\Sigma fx = 256$	Σfx^2

$$\Sigma fx^2 = 1706$$

i)

$$\bar{X}_A = \frac{256}{50} = 5.12$$

$$\sigma = \sqrt{\frac{1706}{50} - \left(\frac{256}{50}\right)^2}$$

$$CVA = \frac{2.81}{5.12} \times 100\% = 54.88\%$$

Model B

x	f	fx	fx^2
1	2	2	2
3	7	21	43
5	12	60	300
7	19	133	931
9	9	81	729
11	1	11	121
$N = 50$		$\Sigma fx = 308$	$\Sigma fx^2 = 2126$

$$i) \bar{x}_B = \frac{308}{50} = 6.16$$

$$s = \sqrt{\frac{2126}{50} - \left(\frac{308}{50}\right)^2} = 2.14$$

$$CV_B = \frac{2.14}{6.16} \times 100\% = 35\%$$

ii) $CV_A > CV_B \therefore A$ is more uniform.

36

	Firm A	Firm B	Total
No of worker	560	500	1060
average monthly wages (\$)	190	200	390
variance	81	100	181
total	831	800	1631

i) As firm B has wages = 200 and firm A has wages = 190. So firm B has larger bill.

$$\begin{aligned}
 \text{ii) } CVA &= \frac{\sigma}{\bar{X}} \times 100\% \\
 &= \frac{\sqrt{81}}{190} \times 100\% \\
 &= 4.74\%
 \end{aligned}$$

$$\begin{aligned}
 CVB &= \frac{\sqrt{100}}{200} \times 100\% \\
 &= 5\%
 \end{aligned}$$

$\therefore$ firm B has greater variability in wage.

combined mean $\bar{X}_{no}$

$$\frac{N_1 \bar{X}_1 + N_2 \bar{X}_2}{N_1 + N_2}$$

$$= \frac{831 \times 190 + 800 \times 200}{1631}$$
$$= 194.90 \text{ \#}$$

combined variance

$$= \frac{N_1 (\sigma_1^2 + d_1^2) + N_2 (\sigma_2^2 + d_2^2)}{N_1 + N_2}$$

$$d_1 = 194.90 - 190 = 4.90$$

$$d_2 = 194.90 - 200 = -5.10$$

$$= \frac{831 (81 + 4.90^2) + 800 (100 + 5.10^2)}{1631}$$

87269.91

x

$$= 115.31 \text{ \#}$$

8

90 no. of orange = N
 $N = 10$

arranging data:

30, 40, 45, 60, 65, 70, 75, 80, 85, 90

smallest value (X_s) = 30

largest " (X_l) = 90

$$\begin{aligned} Q_1 &= \frac{N+1}{4} \\ &= \frac{11}{4} = 2.75^{\text{th}} \text{ term} \\ &= 40 + (45-40) \times 0.75 \\ &= 43.75 \end{aligned}$$

$$\begin{aligned} M_d &= \frac{N+1}{2} = 5.5^{\text{th}} \text{ term} \\ &= \frac{65+70}{2} = 67.50 \end{aligned}$$

$$\begin{aligned} Q_3 &= 3 \times 2.75 = 8.25^{\text{th}} \text{ term} \\ &= 80 + (80-75) \times 0.25 \\ &= 81.25 \end{aligned}$$

i) five summary = (X_s, Q_1, Md, Q_3, X_L)
 $= (30, 43.75, 67.50, 82.5, 90)$

ii)

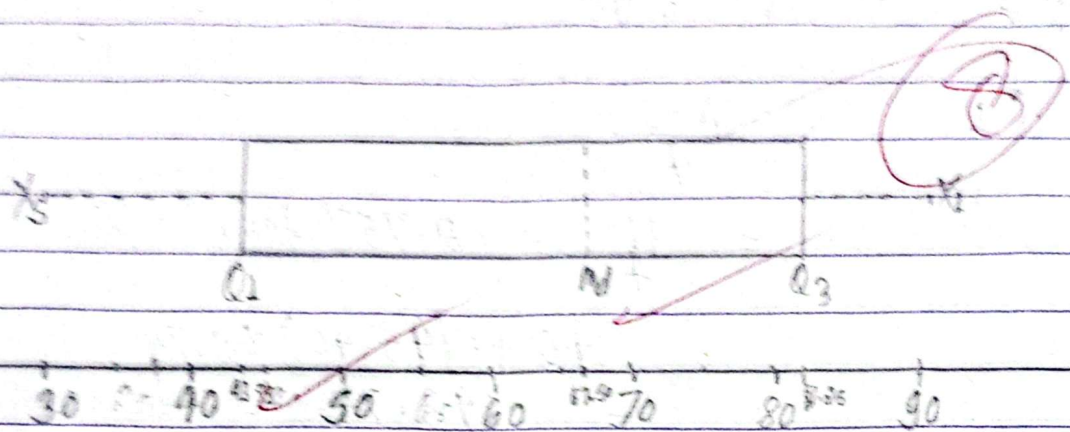


fig: box and whisker plot

$$Md - X_s = 37.50, \quad X_L - Md = 22.50 \quad // \text{ left skewed}$$

$$Md - Q_1 = 23.75, \quad Q_3 - Md = 15 \quad // \text{ left skewed}$$

$$Q_1 - X_s = 13.75, \quad X_L - Q_3 = 7.5 \quad // \text{ left skewed}$$

$\therefore$ the data are left skewed.

40

Statement of Baye's theorem:

If $E_1, E_2, E_3, \dots, E_n$ are mutually exclusive event of statespace S where $P(E_i) \neq 0$ and a arbitrary event A of the space is where $P(A) > 0$ is form as:

$$P(E_r/A) = \frac{P(E_r) \cdot P(A/E_r)}{\sum_{i=1}^n P(E_i) P(A/E_i)}, 1 \leq r \leq n$$

proof: A is subset of S .

$A \subset S$

where A = arbitrary event

S = statespace

E = events

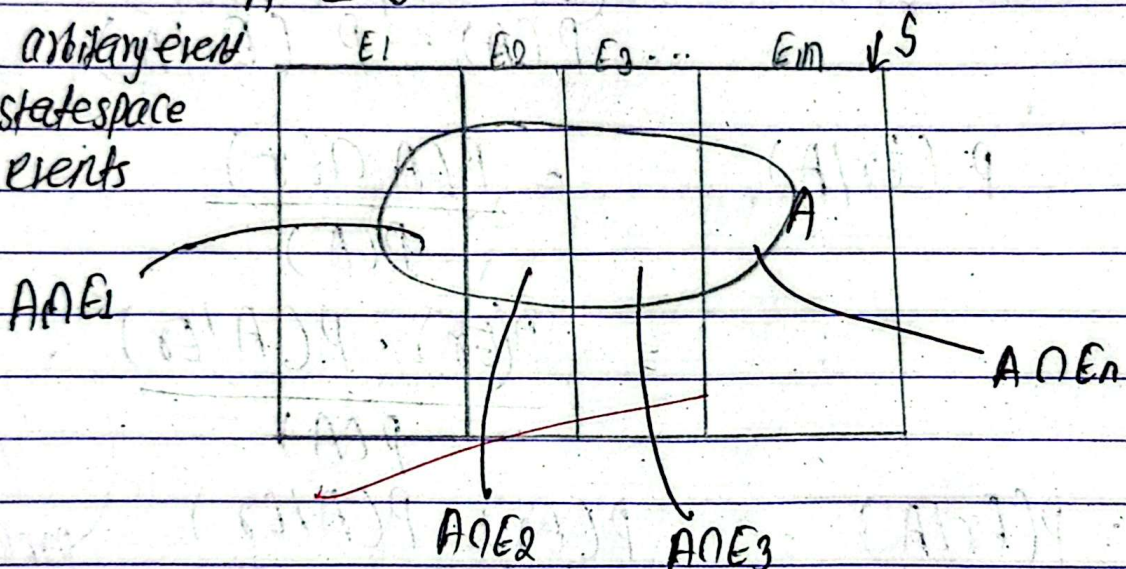


fig: $A \subset S$

we know,

$$S = E_1 \cup E_2 \cup E_3 \cup E_4 \cup \dots \cup E_n$$

$$A = A \cap S \\ = (E_1 \cap A) \cup (E_2 \cap A) \cup (E_3 \cap A) \cup \dots \cup (A \cap E_n)$$

$$P(A) = P(E_1 \cap A) \cup P(E_2 \cap A) \cup P(E_3 \cap A) \cup \dots \cup P(A \cap E_n)$$

$$= P(E_1) \cdot P(A|E_1) + P(E_2) \cdot P(A|E_2) + P(E_3) \cdot P(A|E_3) + \dots + P(E_n) \cdot P(A|E_n)$$

$$\frac{P(E_i \cap A)}{P(A)} = \sum_{i=1}^n P(E_i) P(A|E_i) \quad (\text{events are mutually exclusive})$$

$$P(A \cap E_r) = \frac{P(E_r) \cdot P(A|E_r)}{P(A) \cdot P(E_r|A)}$$

$$P(E_r|A) = \frac{P(A \cap E_r)}{P(A)}$$

$$= \frac{P(E_r) \cdot P(A|E_r)}{P(A)}$$

$$\therefore P(E_r|A) = \frac{P(E_r) \cdot P(A|E_r)}{\sum_{i=1}^n P(E_i) P(A|E_i)}$$



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Dhangadhi, Kailali (Nepal)

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5a

no. of MCQ (n) = 10

options = 4

p (getting correct answer) = $p = 1/4$

q (getting wrong answer) = $q = 3/4$

①

p (getting two questions)

$$P(X=2) = {}^{10}C_2 \left(\frac{1}{4}\right)^2 \left(\frac{3}{4}\right)^8$$

$$= 0.28 \quad \#$$

② p (getting at least one question)

$$= 1 - P(X=0)$$

$$= 1 - {}^{10}C_0 \left(\frac{1}{4}\right)^0 \left(\frac{3}{4}\right)^{10} = 0.94 \quad \#$$

(iii) P (at most two question)

$$= P(X=0) + P(X=1) + P(X=2)$$

$$= 0.06 + {}^{10}C_1 (1/4) (3/4)^9 + 0.28$$

$$= 0.53 \#$$

5b

X	P(X)	$xP(X)$	$x^2 P(X)$
-2	0.1	-0.2	0.4
-1	k	-k	k
0	0.2	0	0
1	2k	2k	2k
2	0.3	0.6	1.2
3	k	3k	9k
total	$0.6+4k$	$0.4+4k$	$12k+1.6$

we know, $\sum P(X) = 1$

$$0.6+4k=1$$

or, $k = 0.10$

$$\sum_x xP(x) = 0.4 + 4 \times 0.2 = 0.80$$

$$\sum_x x^2 P(x) = 2.80$$

$$\text{mean} = 0.80 \#$$

$$\begin{aligned} \text{variance} &= 2.80 - (0.80)^2 \\ &= 2.16 \# \end{aligned}$$

$$\begin{aligned} \text{standard deviation} &= \sqrt{2.16} = \\ &= 1.47 \# \end{aligned}$$

6

6a

X	Y	U = X - A	V = Y - B	U ²	V ²	UV
92	86	15	1	225	1	15
89	88	12	3	144	9	36
86	77	9	-8	81	64	-72
83	68	6	-17	36	289	102
77	85	0	0	0	0	0
71	52	-6	-33	36	1089	-198
68	82	-14	-3	196	9	42
53	37	-24	-48	576	2304	1152
50	57	-27	-28	729	784	756
		$\sum U = -29$	$\sum V = -133$	$\sum U^2 = 2023$	$\sum V^2 = 4549$	$\sum UV = 2229$
$n = 9$						

∴ correlation coefficient =

$$= \frac{n \sum UV - (\sum U)(\sum V)}{\sqrt{n \sum U^2 - (\sum U)^2} \sqrt{n \sum V^2 - (\sum V)^2}}$$

$$= \frac{9 \times 2229 - (29) \times (133)}{\sqrt{9 \times 2023 - 29^2} \sqrt{9 \times 4549 - 133^2}}$$

$$= \frac{9 \times 2229 - 3857}{\sqrt{9 \times 2023 - 841} \sqrt{9 \times 4549 - 17689}}$$

$$= 0.81 \#$$

7

6b

Height (cm) = X
weight (kg) = Y

(Y on X)

N = 8

X	Y	X ²	XY
162	85	26244	10530
168	65	28244	10920
174	84	30276	14616
176	63	30976	11088
180	75	32400	13500
180	76	32400	13680
182	82	33124	14924
184	65	33856	11960
$\Sigma X = 1406$	$\Sigma Y = 575$	$\Sigma X^2 = 247520$	$\Sigma XY = 101218$

regression equation Y on X is:

$$Y = a + bX$$

$$\Sigma Y = na + b \Sigma X$$

$$\textcircled{1} \quad Y = a + bX \quad - \textcircled{A}$$

$$\Sigma Y = na + b \Sigma X$$

$$575 = 8a + 1406b \quad - \textcircled{1}$$

$$\Sigma XY = a \Sigma X + b \Sigma X^2$$

$$101218 = 1406a + 247520b \quad - \textcircled{2}$$

solving ① & ②:

$$a = 3.46$$

$$b = 0.39$$

$$Y = 3.46 + 0.39X$$

, which is required regression equation.

$$\text{ii) } Y(175) = 3.46 + 0.39 \times 175 \\ = 71.71$$

$\therefore$ The weight of man whose height of 175cm is 71.71 Kg.

8

71) Sources of data

Sources of data means the resources from which we get data, fact or information. Data can be any form like text, audio, image, video, document etc. Data is very useful in statistical analysis.

Types of sources of data:

- Primary sources
- Secondary sources

(i) Primary Sources

fact Primary sources are the sources of data from which we get factual data. A observer go on field and collect data himself or herself. For example: data gain from experiment, survey, interview, field visit, questionnaire, seminar etc. In this a data collector and participants interact directly to each other.

(ii) Secondary Sources

These are the sources we get reference data of primary sources. It is the platform in which we get ~~per~~ data that are representation of primary data sources. Example book, newspapers, encyclopedia,

biography, report etc.

Both primary and secondary data sources provide data that are very useful for statistical analysis that help in future plan and research project.

7iii) Properties of correlation coefficient.

→ Correlation shows relation between two variables. Correlation coefficient is factor part of correlation. It is denoted by r .

→ Properties of Correlation Coefficient:

① correlation coefficient lies between 0 and 1.

i.e. $-1 \leq r \leq 1$

② Correlation coefficient is square root of two regression coefficient.



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i.e. $r = \sqrt{b_{yx} b_{xy}}$

where, b_{yx} = regression coefficient of Y on X

b_{xy} = " " " " X on Y.

Correlation

(iii) Regression coefficient can be expression in term of regression coefficient and standard deviation. $r = b_{yx} \frac{\sigma_y}{\sigma_x}$

i.e. $r = b_{yx} \frac{\sigma_y}{\sigma_x}$

$r = b_{xy} \frac{\sigma_x}{\sigma_y}$

$r = b_{yx} \frac{\sigma_y}{\sigma_x}$